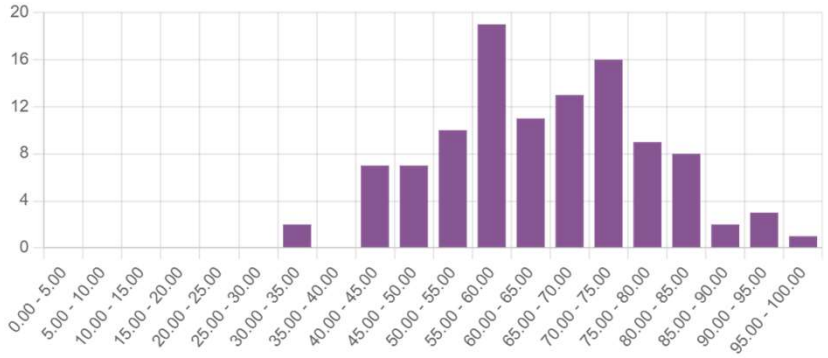
Agenda

- Midterm results
- Midterm question solution

- Regression review
- Estimating the variance of errors
- Estimating b_1 and its distribution

Midterm Grades

Average: 63.49
Standard deviation: 14.75



MSE Question Version 1

Estimator mean MSE B Version 6 (latest)

Question 1
Not yet answered
Marked out of 1.00

Let $\{X_1, X_2, \ldots, X_9\}$ be independent and identically random variables with mean μ and standard deviation σ. We define the following estimators for the mean μ:

$\hat{\theta}_1 = \frac{\sum_{i=1}^{8} X_i}{8}$ and $\hat{\theta}_2 = 0.8X_7 - 0.2X_2$

Choose the correct statement:

- a. The mean squared error of $\hat{\theta}_1$ is always smaller than the mean squared error of $\hat{\theta}_2$.
- b. The mean squared error of $\hat{\theta}_2$ is always smaller than the mean squared error of $\hat{\theta}_1$.
- c. The mean squared error of $\hat{\theta}_1$ and $\hat{\theta}_2$ are always the same.
- d. None of the other statements is correct.

MSE Question Version 2

Estimator mean MSE A Version 5 (latest)

Question 1
Not yet answered
Marked out of 1.00

Let $\{X_1, X_2, \ldots, X_9\}$ be independent and identically random variables with mean μ and standard deviation σ. We define the following estimators for the mean μ:

$\hat{\theta}_1 = \frac{\sum_{i=1}^{8} X_i}{8}$ and $\hat{\theta}_2 = 2X_7 - X_2$

Choose the correct statement:

- a. The mean squared error of $\hat{\theta}_1$ is always smaller than the mean squared error of $\hat{\theta}_2$.
- b. The mean squared error of $\hat{\theta}_1$ and $\hat{\theta}_2$ are always the same.
- c. The mean squared error of $\hat{\theta}_2$ is always smaller than the mean squared error of $\hat{\theta}_1$.
- d. None of the other statements is correct.

Simple linear regression (1)

- The sample: $\{(x_i, y_i)\}_{i=1}^{n}$

independent variable

dependent variable

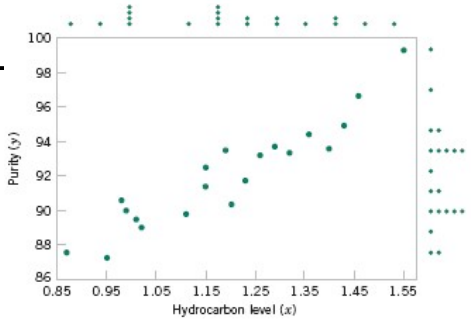
$$y_i = \beta_0 + \beta_1 x_i + \varepsilon$$

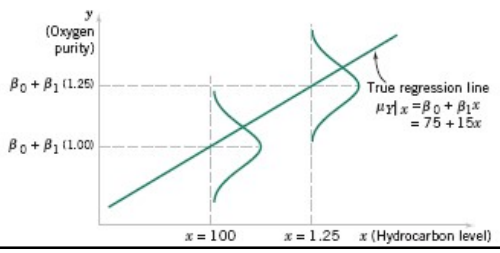
$$\varepsilon \sim N(0, \sigma^2)$$

random deviation or random error

Empirical Models

Observation Number	Hydrocarbon Level x(%)	Purity y(%)
1	0.99	90.01
2	1.02	89.05
3	1.15	91.43
4	1.29	93.74
5	1.46	96.73
6	1.36	94.45
7	0.87	87.59
8	1.23	91.77
9	1.55	99.42
10	1.40	93.65
11	1.19	93.54
12	1.15	92.52
13	0.98	90.56
14	1.01	89.54
15	1.11	89.85
16	1.20	90.39
17	1.26	93.25
18	1.32	93.41
19	1.43	94.98
20	0.95	87.33





$y_i = \beta_0 + \beta_1 x_i + \varepsilon$

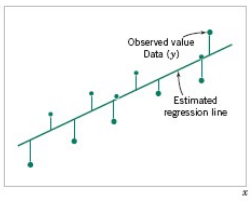
What are we interested in?

Building the model

- Estimating β_0 and β_1

→ the fitted regression line

b_0, b_1: Minimize the sum of squared deviations.



- Estimating σ
- Distribution of $b_0, b_1, \hat{Y} = \hat{\mu}_{Y|x_0} = b_0 + b_1 x_0$

$$b_0 \sim N\left(\beta_0, \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)\right)$$

Evaluating the model

- Is the model significant?
- How useful is the model?

Estimating σ^2 (1)

$$\varepsilon_i \sim N(0, \sigma^2)$$

variability of each of Y_i about the true regression line.

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

variability of data pts about the fitted regression line.

$$\hat{\sigma}^2 = ?$$

Estimating σ^2 (2)

$two \quad parameters \quad to \quad estimate \quad \beta_0, \beta_1$

$$E[SSE] = (n-2)\sigma^2 \quad \Rightarrow \quad \hat{\sigma}^2 = \frac{SSE}{n-2} = MSE$$

can be shown

Two parameters to Estimate β_0, β_1 $\Rightarrow$ a loss of 2 df

MSE is an unbiased estimator for σ^2.

Example: Oxygen Purity

Regression Analysis

The regression equation is

Purity = 74.3 + 14.9 HC Level

Predictor	Coef	SE Coef	T	P
Constant	74.283 ← $\hat{\beta}_0$	1.593	46.62	0.000
HC Level	14.947 ← $\hat{\beta}_1$	1.317	11.35	0.000

S = 1.087 R-Sq = 87.7% R-Sq (adj) = 87.1%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	152.13	152.13	128.86	0.000
Residual Error	18	21.25 ← SS_E	1.18 ← $\hat{\sigma}^2$		
Total	19	173.38			

Predicted Values for New Observations

New Obs	Fit	SE Fit	95.0% CI	95.0% PI
1	89.231	0.354	(88.486, 89.975)	(86.830, 91.632)

Values of Predictors for New Observations

New Obs	HC Level
1	1.00

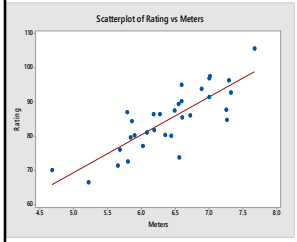
Example: Python Output

	df	sum_sq	mean_sq	F	PR(>F)
X	1.0	152.13	152.13	128.86	0.000
Residual	18.0	21.25	1.18	NaN	NaN

SSE MSE

Class Exercise

A data set on the ratings of quarterbacks for the 2008 National Football League season is collected by The Sports Network. The rating (y) is suspected to be related to the average number of meters gained per pass attempt (x).



Regression Analysis: Rating versus Meters

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
Regression	1	1669.02	1669.02	61.02	0.000
Error	30	820.50	27.35		
Total	31	2489.52			

Model Summary

S	R-sq	R-sq(adj)	R-sq(pred)
5.22972	67.04%	65.94%	62.11%

Coefficients

Term	Coef	SE Coef	T-Value	P-Value
Constant	14.19	9.09	1.56	0.129
Meters	11.04	1.41	7.81	0.000

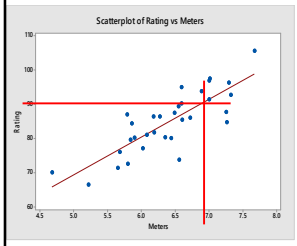
Regression Equation

Rating = 14.19 + 11.04 Meters

What is the estimate of σ^2?

Class Exercise

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Regression Analysis: Rating versus Meters

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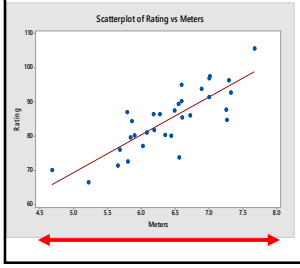
90.366

Find an estimate of the mean rating if a quarterback averages 6.9 m per attempt.

32

Class Exercise

A data set on the ratings of quarterbacks for the 2008 National Football League season is collected by The Sports Network. The rating (y) is suspected to be related to the average number of meters gained per pass attempt (x).



Regression Analysis: Rating versus Meters

Analysis of Variance

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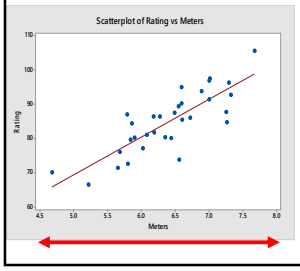
-9.936

What change in the mean rating is associated with a decrease of 0.9 m per attempt?

33

Class Exercise

A data set on the ratings of quarterbacks for the 2008 National Football League season is collected by The Sports Network. The rating (y) is suspected to be related to the average number of meters gained per pass attempt (x).



Regression Analysis: Rating versus Meters

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Meters	11.04	1.41	7.81	0.000

Regression Equation

Rating = 14.19 + 11.04 Meters

15/11.04=1.359

To increase the mean rating by 15 points, how much must an increase in the average m per attempt be generated?

Class Exercise

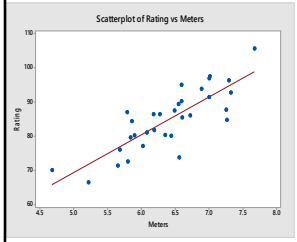
86.9436

A data set on the ratings of quarterbacks for the 2008 National Football League season is collected by The Sports Network. The rating (y) is suspected to be related to the average number of meters gained per pass attempt (x).

Regression Equation

Rating = 14.19 + 11.04 Meters

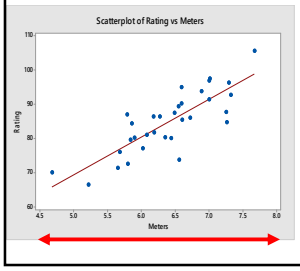
Meter	Rating	Meter	Rating	Meter	Rating	Meter	Rating
7.67	105.5	6.59	90.2	6.6	85.4	5.84	79.6
7.01	97.4	6.55	89.4	7.26	84.7	6.02	77.1
7	96.9	7.25	87.7	5.86	84.3	5.68	76
7.29	96.2	6.49	87.5	6.19	81.7	6.56	73.7
6.59	95	5.79	87	6.08	81	5.8	72.6
6.89	93.8	6.18	86.4	6.35	80.3	5.65	71.4
7.32	92.7	6.27	86.4	5.9	80.2	4.68	70
7	91.4	6.72	86	6.44	80.1	5.22	66.5



Given that x = 6.59 m, find the fitted value of y and the corresponding residual.

Class Exercise

A data set on the ratings of quarterbacks for the 2008 National Football League season is collected by The Sports Network. The rating (y) is suspected to be related to the average number of meters gained per pass attempt (x).



Regression Analysis: Rating versus Meters

Analysis of Variance

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Coefficients

Term	Coef	SE Coef	T-Value	P-Value
Constant	14.19	9.09	1.56	0.129
Meters	11.04	1.41	7.81	0.000

Regression Equation

Rating = 14.19 + 11.04 Meters

Using this regression model, can you find an estimate for x = 9 m? Why or why not? If you can, find the fitted value of y.

Inferences about slope, β_1 (1)

- If β_1 =0, what is the meaning of regression?
- $b_1 = \hat{\beta}_1 \Rightarrow$ inferences for β_1 through analysis of b_1.
- **Distribution of b_1:**

$$b_1 = \frac{S_{xy}}{S_{xx}} \quad \Rightarrow b_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$$

But σ^2/S_{xx} is not known...

Inferences about slope, β_1 (2)

σ^2 is not known, but can be estimated with df=n-2:

$$\hat{V}(b_1) = \hat{\sigma}_{b_1}^2 = s_{b_1}^2 = \frac{\hat{\sigma}^2}{S_{xx}} \quad \Rightarrow \quad s_{b_1} = \frac{\hat{\sigma}}{\sqrt{S_{xx}}}$$

$$\Rightarrow \quad T = \frac{b_1 - \beta_1}{\hat{\sigma}/\sqrt{S_{xx}}} = \frac{b_1 - \beta_1}{S_{b_1}} \sim t_{n-2}$$

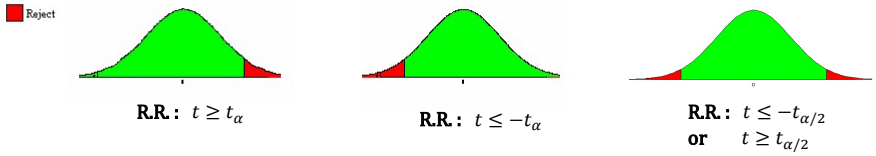
Confidence interval for β_1

- Straightforward when the distribution of β_1 is known as t_{n-2}…
  - Your book has the details …

Hypothesis testing on β_1

$H_0: \beta_1 = \beta_1^0$ Test statistics: $T = \dfrac{b_1 - \beta_1^0}{s_{b_1}} \sim t_{n-2}$

$H_a: \beta_1 > \beta_1^0$ $H_a: \beta_1 \neq \beta_1^0$ $H_a: \beta_1 < \beta_1^0$



R.R.: $t \geq t_\alpha$ R.R.: $t \leq -t_\alpha$ R.R.: $t \leq -t_{\alpha/2}$ or $t \geq t_{\alpha/2}$

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Significance of regression analysis

$H_0: \beta_1 = 0$

$H_a: \beta_1 \neq 0$

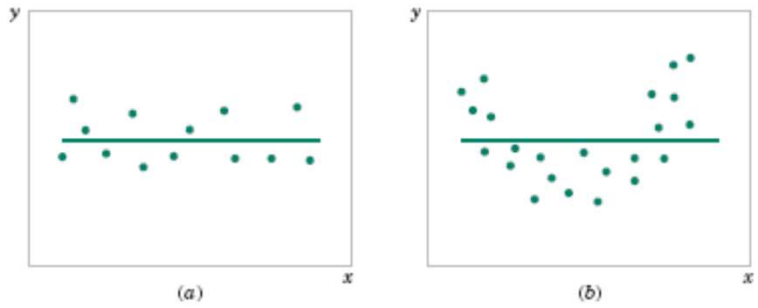
Test statistics: $T = \dfrac{b_1}{s_{b_1}} \sim t_{n-2}$



R.R.: $t \leq -t_{\alpha/2}$ or $t \geq t_{\alpha/2}$

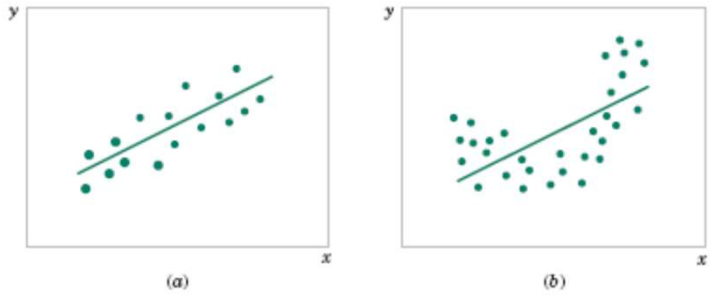
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Hypothesis Tests in Simple Linear Regression



The hypothesis H_0: $\beta_1 = 0$ is not rejected.

Hypothesis Tests in Simple Linear Regression



The hypothesis H_0: $\beta_1 = 0$ is rejected.

Example: Oxygen Purity

Regression Analysis

The regression equation is

Purity = 74.3 + 14.9 HC Level

Predictor	Coef	SE Coef	T	P
Constant	74.283 ← $\hat{\beta}_0$	1.593	46.62	0.000
HC Level	14.947 ← $\hat{\beta}_1$	1.317	11.35	0.000

$$\frac{\hat{\beta}_1}{\sqrt{\hat{\sigma}^2/S_{xx}}} = \frac{\hat{\beta}_1}{se(\hat{\beta}_1)}$$

S = 1.087 R-Sq = 87.7% R-Sq (adj) = 87.1%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	152.13	152.13	128.86	0.000
Residual Error	18	21.25 ← SS_E	1.18 ← $\hat{\sigma}^2$		
Total	19	173.38			

Predicted Values for New Observations

New Obs	Fit	SE Fit	95.0% CI	95.0% PI
1	89.231	0.354	(88.486, 89.975)	(86.830, 91.632)

Values of Predictors for New Observations

New Obs	HC Level
1	1.00

ANalysis Of VAriance identity

$$SST = S_{yy} = \sum (y_i - \overline{y})^2 = \sum y_i^2 - \frac{\sum y_i^2}{n}$$

total variability of Y.

$$SSE = \sum (y_i - \hat{y}_i)^2$$

error sum of squares
Interpretation: unexpected variability of Y in regression.

$$SSR = \sum (\hat{y}_i - \bar{y})^2$$

regression sum of squares
Interpretation: amount of variability that is explained by the model.

$$SST = \sum (y_i - \overline{y})^2 = \sum (\hat{y}_i - \overline{y})^2 + \sum (y_i - \hat{y}_i)^2$$

df=n-1 df=1 df=n-2

ANalysis Of VAriance identity

$E(SSE/(n-2)) = \sigma^2$ $\quad$ $SSE/\sigma^2 \sim \chi^2_{n-2}$

$E(SSR) = \sigma^2 + \beta_1^2 S_{xx}$ $\quad$ $SSR/\sigma^2 \sim \chi^2_1$

If β_1=0, what is the distribution of $\frac{SSR/1}{SSE/(n-2)}$?

$\frac{W/u}{Y/v} \sim F_{u,v}$ $\quad$ $W \sim \chi^2_u$, $Y \sim \chi^2_v$

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ANalysis Of Variance (ANOVA) table for simple regression

Source	*df*	*Sum Sq.*	*Mean Sum Sq.*	*F*
Regression	*1*	*SSR*	$MSR = \frac{SSR}{1} = SSR$	$\frac{SSR}{MSE}$
Error	*n-2*	*SSE*	$MSE = \hat{\sigma}^2 = \frac{SSE}{n-2}$	
Total	*n-1*	*SST*		

$H_0: \beta_1 = 0$

$H_a: \beta_1 \neq 0$

→ can be tested via:

$t^2 = f \quad \Rightarrow \quad t^2_{\alpha/2,n-2} = F_{\alpha,1,n-2}$

reject H_0 if $F > F_{\alpha,1,n-2}$.

↓

equivalent to t test in simple linear regression. F-test is only two-sided.

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Is the linear regression model significant?

$H_0: \beta_1 = 0 \qquad H_a: \beta_1 \neq 0$

Test statistics:

$$T = \frac{b_1}{s_{b_1}} \sim t_{n-2}$$



R.R.: $t \leq -t_{\alpha/2}$
or $t \geq t_{\alpha/2}$

$t^2 = f \quad \Rightarrow \quad t^2_{\alpha/2,n-2} = F_{\alpha,1,n-2}$
reject H_0 if $F > F_{\alpha,1,n-2}$.

only two-sided